

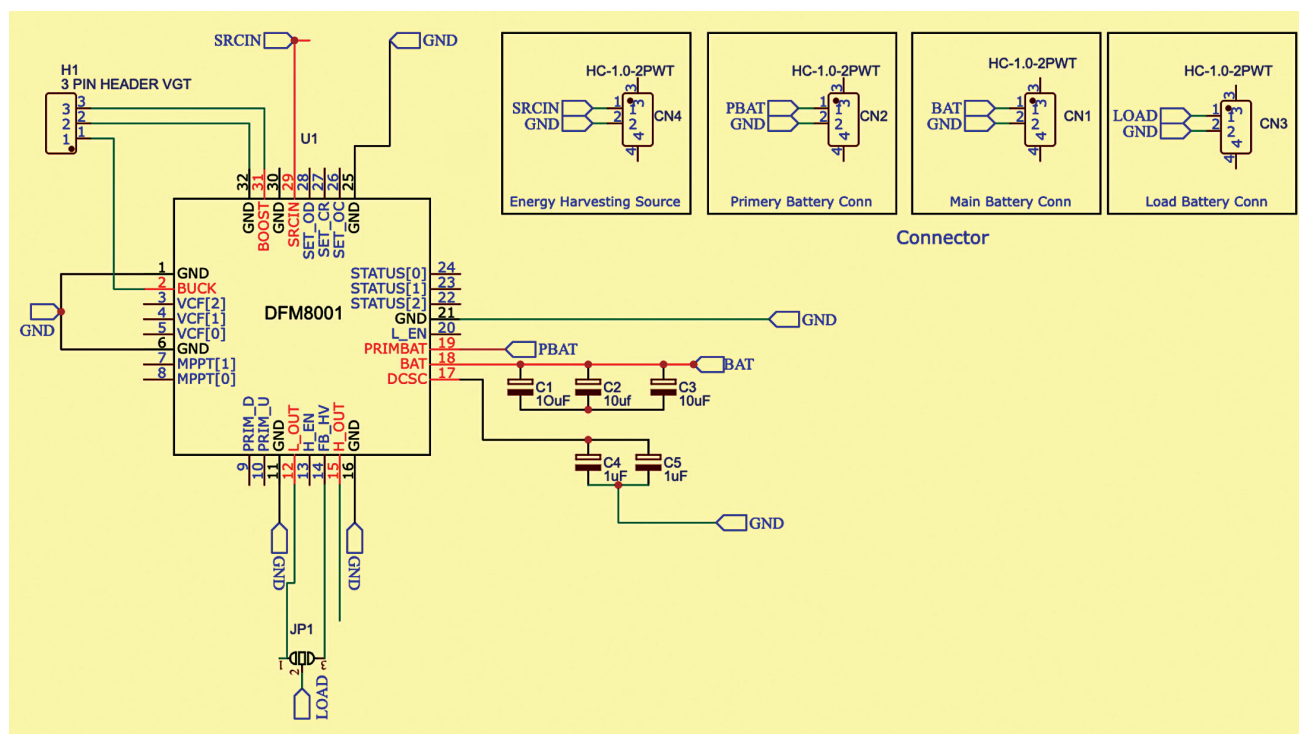


Fig. 2: Bare-minimum circuit of the DFM8001 energy-harvesting system

2. The LTC3588 energy-harvesting system

DFM8001 energy harvesting power system

Fig. 2 shows the bare-minimum circuit of the module. The presented circuit diagram is a simplified version, with unnecessary chip components and functions removed and only essential ones retained. It is built around the DFM8001 IC, supercapacitors, and a few additional components.

This design combines a supercapacitor that stores harvested energy with the ability to power very low-power IoT devices, or it can be paired with a battery to extend the charge and power cycle. According to the DFM8001 datasheet, the system can harvest energy from solar cells, thermal energy from a thermoelectric generator, or radio waves. The chip supports multiple operating modes and includes an option to integrate a supercapacitor to store harvested energy and supply power to the load.

The datasheet specifies that the

chip features an integrated dual LDO regulator with low-voltage outputs of 1.2V and 1.8V at 20mA, and high-voltage outputs from 1.8V to 4.1V at 80mA. It also offers dynamic control over LDO output activation and highly flexible energy-storage management. This includes adjustable overcharge and overdischarge protection, compatibility with any rechargeable battery or capacitor, a battery-low warning, and LDO output support for battery backup. When the rechargeable battery is depleted, the system automatically switches to the backup battery after 600 milliseconds.

Battery configuration. The DFM8001 supports flexible multi-battery configurations for low-power IoT devices through three dedicated connectors. The primary battery input powers the system when harvested energy is unavailable, while the main battery connector stores energy in a rechargeable cell or supercapacitor. The load connector supplies stable power to the application, preventing deep

discharge. Capacitors (C1-C5) on the BAT and PRIMBAT lines smooth charge and discharge cycles, reduce noise, and provide short bursts during periods of peak load demand.

Pinouts and application of DFM8001 IC. The DFM8001 is a highly integrated energy-harvesting IC designed to extract power from sources such as photovoltaic cells, vibration harvesters, and other micro-energy inputs. The left side of the IC provides connections for energy input and configuration, including buck and boost outputs, VCF configuration pins, MPPT voltage tracking input, and control settings for over-discharge, charging rate, and over-current protection. The right side of the chip mainly handles power storage and system interfaces, offering dedicated pins for primary and secondary (Li-ion or rechargeable) batteries, load supply rails (H_OUT and L_OUT), high-voltage feedback, and status-monitoring outputs.

In the typical application diagram, the DFM8001 manages